Kirolos Sedra

Associate Applications Engineer | RF, Lab Validation & Test Automation

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Professional Summary

Applications engineer with an MASc in Electrical & Computer Engineering and hands-on experience reproducing system failures in the lab, designing controlled experiments, characterizing RF/wireless behavior, automating test and provisioning workflows, and communicating results to customers and engineering partners. Performed VNA measurements for S11, coupling, and crosstalk; built TinySA-based spectrum/power measurement pipelines; and debugged systems across firmware, RF, networking, and cloud boundaries. Delivered customer demonstrations, trained engineers on reproducible test procedures, supported a 50-gateway deployment, and presented technical findings to Apple, Rogers, telecom partners, and mixed technical audiences.

Core Applications Engineering Skills

Customer / Applications: issue reproduction, root-cause analysis, requirements discovery, qualification/validation support, technical demos, training, SOPs, stakeholder presentations, vendor/support coordination
RF / Signal Integrity / Lab: VNA measurements (S11, coupling, crosstalk), TinySA spectrum analysis, USRP B210 SDR, RF attenuators, anechoic-chamber testing, RSSI/SNR/SINR/RSRP/RSRQ measurements
Test / Automation: Python, automated data collection, test orchestration, hardware-triggered E2E testing, device provisioning, post-processing, visualization, real-time dashboards, repeatable validation flows
Embedded / Wireless: C/C++, BLE, LTE/5G, Wi-Fi 6/6E, MQTT, nRF52/nRF91, ESP32, Raspberry Pi, Quectel UE, Zephyr/Nordic SDK
Networking / Integration: TCP/IP, DNS, DHCP, RADIUS, SSH, Wireshark, tcpdump, ArubaOS/AirRecorder, AT/QMI, REST APIs, SQL, Azure IoT Hub, Azure CLI, Git, Docker, Linux

Experience

AquaSensing **Sep 2024 – Present**
Research Intern → Firmware Specialist / Product Engineer *Waterloo, ON*

- Worked across the complete smart-IoT path—sensor → BLE → gateway → LTE → MQTT → cloud → notification—reproducing and resolving failures across firmware, connectivity, backend, cloud, and user-facing behavior.
- Prepared and ran an end-to-end demonstration for **12 Energy** using real sensors and gateways; the customer subsequently requested **50 gateways**. Helped provision the fleet across firmware, Azure IoT Hub identities/credentials, and SQL records while automating repeated device registration with **Azure CLI** and supporting deployment follow-up.
- Led SDK migration testing and rollout; reproduced an **LTE reconnection failure** by forcing signal loss in an anechoic chamber, traced it to build configuration, validated recovery, and ran carrier experiments including a **10-SIM Bell/Jasper trial** before rejecting a migration that the evidence did not support.
- Replaced a manual daily full-chain smoke test with an **ESP32 automated E2E test kit**, reusable subsystem stubs, and SOP **AQS-BLE-002**; trained **five engineers** so validation was reproducible and transferable rather than person-dependent.

University of Waterloo – Apple & Rogers Collaborations **Sep 2024 – Aug 2026**
Graduate Researcher, Wireless Sensors and Devices Lab *Waterloo, ON*

- Built and automated production-representative Wi-Fi test environments across Aruba, ASUS, and TP-Link infrastructure, combining DNS/DHCP/RADIUS services, SSH-managed devices, logging, RF instrumentation, and custom orchestration; reduced manual intervention by approximately **70%**.
- Built a **Raspberry Pi + TinySA** RF measurement pipeline for repeatable spectrum/power collection and frequency-time waterfall analysis; performed **VNA measurements for S11, coupling, and crosstalk** when characterizing RF behavior.
- Challenged an apparent multi-second throughput ramp by questioning measurement granularity; used **tcpdump** to separate real transfer behavior from a 1 Hz reporting artifact and debugged across DUT, packet capture, RADIUS, Aruba OS/AirRecorder, WLAN infrastructure, and direct vendor support.
- Presented progress weekly to **Apple and Rogers** stakeholders; built live-data views for mixed technical audiences and demonstrated a three-station BLE localization system using nRF52 receivers, directional antennas, firmware RSSI thresholds, and a real-time GUI to Apple executives.
- Built/debugged a private **5G SA** environment using srsRAN, Open5GS, USRP B210, and Quectel UE; diagnosed registration/session issues and characterized latency, throughput, mobility, and **RSRP/RSRQ/SINR, MCS, and CQI**.

Ejada Systems **Sep 2023 – Aug 2024**
Software Engineer *Full-time, Hybrid*

- Delivered customer-specific Java/Spring banking APIs for **Al Rajhi Bank** inside a Newgen BPM → IBM ESB/middleware → backend → Microsoft SQL Server environment, progressing changes through Dev, SIT with customer data, and Production while coordinating with middleware teams and a bank-side consultant.
- Debugged and refactored legacy services, improved measured SQL/data-access performance by approximately **30%**, and helped establish a greenfield Emkan BNPL backend in a two-person core team with clear application and integration boundaries.

University of Waterloo **Feb 2025 – Apr 2026**
Teaching Assistant *Part-time, Contract*

- Taught tutorials/labs, held office hours, and supported student troubleshooting for ECE 106, ECE 610 (networking/Mininet, two terms), and ECE 358; translated technical concepts into clear procedures and feedback for mixed skill levels.

Education

University of Waterloo **Sep 2024 – Aug 2026**
MASc, Electrical & Computer Engineering – Pattern Analysis & Machine Intelligence *92% (A+)*
Thesis: Experimental comparison of Wi-Fi 6 and 5G NSA/SA for mobile-robot communications; two peer-reviewed CCECE 2026 papers and conference presentations.

Ain Shams University **Sep 2018 – Jul 2023**
B.Sc., Computer & Systems Engineering *CGPA 3.12/4.00*
Graduation project: digital-twin self-driving system using Siemens Simcenter PreScan with RADAR, LiDAR, camera sensing, sensor fusion, and ROS integration/testing; project grade 4.00/4.00.

Selected Publications & Technical Communication

- CCECE 2026 – *Scalability Analysis of Wi-Fi Broadcast Mechanisms Using Simulation and Hybrid Approaches.*
- CCECE 2026 – *Trajectory-Based Performance Characterization of Wi-Fi for Outdoor Mobile Robot Teleoperation.*
- Technical communication includes SOPs, validation procedures, conference presentations, executive demos, customer/partner discussions, training, and evidence-based deployment recommendations.